

How temperature and substrate affect anemone boldness.pdf

by Yuki Wu

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Coursework Coversheet

Name	Xiaoqi (Yuki) Wu
Date	14/06/2024
Module	Ecological field skills
Coursework item	Mini-project

Self-reflection is a non-assessed part of the coursework submission to aid your learning

1. Have you used previous feedback/self-reflection to produce this coursework? If so, how? If not, why?

Yes, I utilized my previous feedback to improve my time management by creating a well-organized schedule for completing this project.

2. What do you feel went well? (e.g., development of specific skills, sense of achievement, part of the assignment you are most proud of)

I did well with time management. I made a point to complete one section every day, allowing ample time for essay revisions.

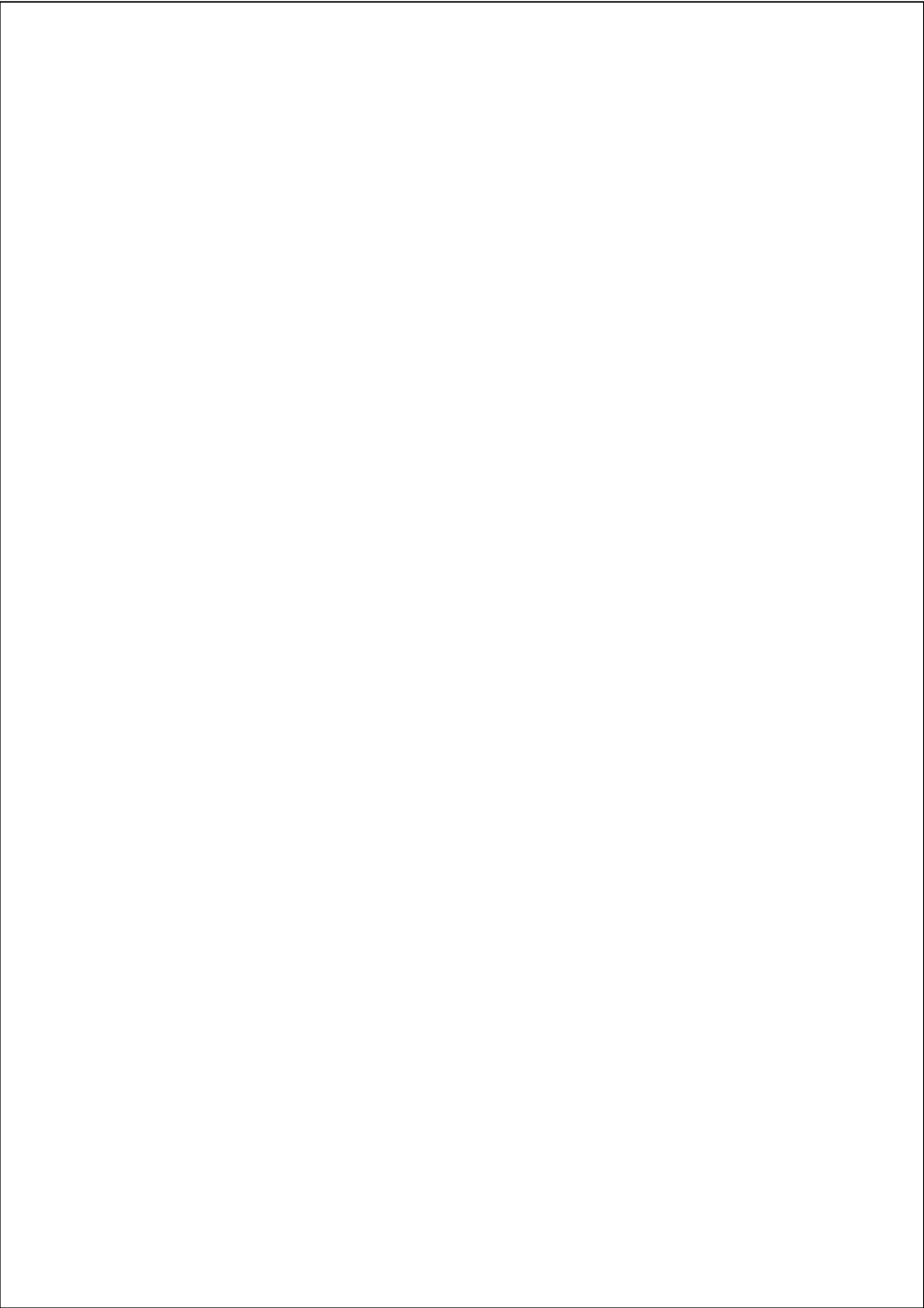
3. What do you feel did not go well? (e.g., learning outcomes that you found most difficult to meet, aspects of the assignment that were challenging, time-management)

What I found challenging was finding related papers to discuss, as there were limited papers had conducted experiments to investigate the relationship between temperature and anemone's behaviours and no paper had conducted about how the presence of mussels affect anemone's behaviours. Therefore, I discussed more on the limitations of our experiment.

Additionally, I discovered that it is quite difficult to make an experiment perfect, given the multitude of variables that need to be considered and controlled.

4. What did you hope to learn from doing this coursework? Did you achieve that?

I anticipated learning about the considerations necessary for experiment design. Although designing and completing the experiment was challenging under limited time and experimental equipment, we still successfully completed our coursework and I really enjoy investigating marine organisms with my teammates. *great*



Investigation of boldness in Sea Anemone (*Metridium senile*) with and without Substrate under Different Temperature Conditions

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ABSTRACT

start with the background/problem before moving on to the study

This study investigates the boldness of the sea anemone (*Metridium senile*) under different temperature conditions and with or without substrates. Anemones were collected from Plymouth Marine Station and divided into groups based on temperature ($14.5 \pm 0.5^\circ\text{C}$ and $19.5 \pm 0.5^\circ\text{C}$) and substrate presence (with or without mussels). *Boldness, measured as* Startle response time (SRT), the time taken for anemones' tentacles to extend to their collar after a fright, was recorded after simulated predatory stimulation by water squirts. Statistical analysis showed that temperature plays a crucial role in anemones' boldness. Compared to anemones in lower temperatures, those in higher temperatures showed significantly shorter SRT, potentially due to enhanced metabolic rates. However, living on mussels did not impact the anemones' boldness. The results contribute to our understanding of how environmental changes, possibly caused by climate change and boat traffic pollution, impact marine organisms and highlight the importance of considering both temperature and habitat conditions in ecological studies. Further research is recommended to explore long-term effects and validate these findings in natural habitats.

KEYWORDS: Sea anemone, startle response time, boldness, temperature conditions, substrate presence

INTRODUCTION

As climate change accelerates, environmental temperatures are undergoing notable changes, such as increased sea surface temperatures by 0.13°C (Masson-Delmotte, 2021). In coastal regions, this warming is exacerbated by pollutants such as heavy metals and nutrients released from boat traffic, further altering seawater temperature and affecting pH levels (Duarte et al., 2013). These fluctuations in temperature and pH profoundly impact the behaviour and physiology of marine organisms, including sea anemones (Putnam & Edmunds, 2011; Maskrey et al., 2021).

Sea anemones are sessile organisms that respond sensitively to environmental changes. Their physiological and behavioural responses, such as growth rate, regeneration rate (Putnam & Edmunds,

3

2011) and change in boldness (Maskrey, 2021), to temperature shifts are critical indicators of how marine ecosystems might adapt to global warming (Putnam & Edmunds, 2011; Kroeker et al., 2010).

4

We chose to investigate *Metridium senile* because it is a widely distributed species found in a variety of marine environments, making it an ideal model organism for studying the impacts of temperature changes on marine life (Sebens, 1984).

In the natural habitat, anemones tend to grow on substrates like mussels, because anemones can acquire stable habitat and food resources by eating mussel larvae. The association between the anemones and their habitat conditions may affect their behaviour if they are under environmental stress, including temperature change (Helmuth et al., 2002).

good to see, but as you can see the reader wants this info earlier if they are a non expert

Previous studies have shown that raising temperature might significantly impact an anemone's boldness behaviour (Maskrey et al., 2020), where boldness is defined as an animal's inclination to take risks and bolder animals will show riskier behaviour than shy animals (Beckmann & Biro, 2013). The boldness of anemones is measured through the startle response time (SRT), which is the latency for the anemone to re-extend its feeding tentacle and surpass 75% of its collar after disturbance (Rudin & Briffa, 2011).

It is meaningful to investigate anemone boldness because boldness can influence how anemones interact with their environment, including their foraging strategies and predator avoidance mechanisms. Their tentacle extension is correlated to both foraging efficiency and metabolic processes, where their metabolic rate reaches the highest when they fully extend their tentacles (Griffiths, 1977). Anemones have to balance between the risk of being predated and growth through foraging and maximising their metabolic rate (Maskrey et al., 2021). However, these studies often overlook the impact of different living conditions, such as growing on mussels, on anemone behaviour under temperature stress, and almost all studies around this topic were around the beadlet anemone (*Actinia equina*).

good

Our study aims to fill this gap by investigating the boldness behaviour of *Metridium senile* in response to temperature changes, considering both substrate-attached and substrate-free conditions. We hypothesize that higher temperatures will increase the boldness of anemones, giving shorter SRT. Additionally, we expect that anemones growing on mussels will be bolder compared to those without a substrate due to the more stable and resource-rich environment provided by mussels (Shick, 2012).

In our experiment, we simulated a predation stimulus by squirting anemones with water and measuring their SRT. By comparing anemones reared at different temperatures and living conditions, we aim to provide insights into their adaptability and the potential implications for marine ecosystem functions under climate change scenarios.

METHODS

Experiment procedure and data collection

water temp?

Metridium senile were randomly collected using a catching net and buckets along the side of the pontoon at Plymouth marine station (N 50 21 50, N 04 07 51). Due to ethical considerations, we limited our collection to 52 anemones, with half growing on mussels and the other half directly on the pontoon. The sizes of anemones were measured and labelled after collection and they were put into separate aquariums with oxygen provided. Given the size variability, the anemones were randomly divided into 2 groups, A and B, ensuring a representative size range in each group. Each group consisted of 26 anemones, with 13 anemones growing on mussels and 13 anemones not growing on mussels. Two groups of anemones were then reared overnight in separate laboratories with controlled environmental temperatures of $14.5 \pm 0.5^\circ\text{C}$ and $19.5 \pm 0.5^\circ\text{C}$, respectively. This acclimation period allowed the anemones to recover from the collection stress and adapt to the laboratory conditions.

On the second day, each anemone in groups A and B was subjected to a standardized water squirt test. A 20ml syringe, held 1 cm above the anemones' mouthparts, was used to squirt 20ml of water onto each anemone. Timing started immediately when water was squirted. The syringe was then refilled, and another 20ml of water was squirted within 10 seconds. The SRT values were recorded. If the anemones have not re-opened to reach 75% of its collar in 50 minutes, it was recorded at 3000 seconds. This observation time was set based on our preliminary observation.

Two squirts of 20ml water were used based on findings by Maskrey et al. (2021), who utilized a 50ml syringe in their experiments with beadlet sea anemones. Preliminary experiments indicated that a single 20ml squirt was insufficient to stimulate a response in *Metridium senile*. The dual stimulation was designed to simulate anti-predator behaviour, with the first squirt causing body contraction and increased internal pressure, and the second squirt inducing a defensive state and ejection.

Following initial testing, the anemones were swapped between the two temperature conditions in a paired test to eliminate order effects. The anemones were then allowed to acclimate for 90 minutes to ensure they were fully reopened before repeating the stimulation and recovery measurement process.

repeatability of SRT to test for boldness personality?

Statistical analysis

R version 4.3.0 was used for all statistical analyses. To ensure the reliability of our observations among group members, we tested inter-reliability by jointly observing a set of anemones and analyzing the data. the intra-class correlations (ICC) test. The ICC value obtained was 0.949, indicating high reliability among different observers. Given the reliability of our observed data, we split into groups to observe the anemone's behaviours due to time constraints the following day.

To assess the impact of different temperatures on anemone recovery behaviour, Wilcoxon signed-rank test was used to compare SRT between low and high-temperature conditions and the effect of substrate presence (with or without mussels) on the different SRT values between high and low

what about the substrate analysis?



temperatures. Model analysis using the Q-Q plot indicated that residuals are not normally distributed. Therefore, using the Wilcoxon test is appropriate as it is suitable for paired data that do not follow a normal distribution. Box plots were generated to visualize the distribution of recovery times under different temperature conditions and substrate types.

Ethical considerations

good

All study animals were ethically sourced from the Plymouth Marine Station and reared specifically for study purposes. No regulatory approval was required due to minimal risk. Throughout the experiment, the welfare of the anemones was ensured through stress minimization, suitable nutrition and living conditions under controlled laboratory conditions to simulate their natural habitat. The simulation of predation stimulus by squirting water was a method chosen to minimize impact. The anemones were immediately returned to their natural environment after the experiment.

RESULTS

The Wilcoxon signed-rank test was used to compare the SRT values of anemones between low ($14.5 \pm 0.5^{\circ}\text{C}$) and high ($19.5 \pm 0.5^{\circ}\text{C}$) temperature conditions. The result showed that there is a significant difference between the low and elevated temperatures ($V = 670.5$, $p\text{-value} = 0.01737$; Table 1). However, no significant difference is shown in the effect of substrate presence on the recovery times of anemones under different temperature conditions ($W = 310$, $p\text{-value} = 0.6138$; Table 1).

Table 1. Summary of Wilcoxon signed-rank test results comparing SRT value for *Metridium senile* anemones between different temperature conditions and substrate types.

Comparison	p-value	Mean Difference (s)	Median Difference (s)
SRT (Low vs. High Temperature)	0.01737 *	-558	-120
SRT (With vs. without Mussels)	0.6138	-94	-14
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1			

p-value is usually the final column

The box plot in Figure 1 visualized the distribution of SRT. The anemones exposed to higher temperatures have shorter median SRTs, indicating faster recovery times (Fig. 1a), along with a more compact interquartile range (IQR), showing less variability in SRT. Figure 1b shows no significant difference in SRT between anemones with and without mussels. For both substrate conditions, the median SRT difference is close to zero and they have similar variability. These visualizations support the conclusion that temperature significantly affects the SRT of *Metridium senile*, while the presence of mussels does not.

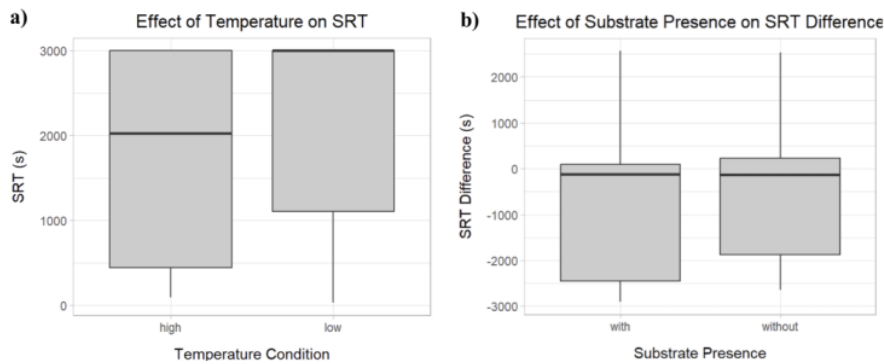


Fig. 1. Box plots illustrated the distribution or difference of SRT for *Metridium senile* anemones under different conditions. **a)** The distribution of SRT in response to low ($14.5 \pm 0.5^\circ\text{C}$) and high ($19.5 \pm 0.5^\circ\text{C}$) temperature. **b)** The effect of substrate presence (with or without mussels) on the difference in SRT between low ($14.5 \pm 0.5^\circ\text{C}$) and high ($19.5 \pm 0.5^\circ\text{C}$) temperatures.

DISCUSSION

Our findings indicate a significant difference in SRT between low and high temperature conditions, which indicates anemones exposed to higher temperatures are bolder. Although SRT can be used as an indicator of measuring boldness, it is also dependent on anemone's metabolic rates. Elevated temperatures may enhance metabolic rates, leading to more rapid physiological responses (Chomsky et al., 2004). Maskrey et al. (2024) found that bolder *Actinia equina* anemones tend to have the highest metabolic rate at high temperatures (21°C). This implies that boldness is affected by temperature-induced metabolic changes. This is consistent with previous studies that have shown increased metabolic activity in marine invertebrates at higher temperatures (Fitt et al., 2000; Ove & John, 2010). In contrast, Maskrey et al. (2020) indicated that *Actinia equina* shows an increase in SRT under higher temperature conditions. This may be due to different species having various responses.

The smaller SRT values at higher temperatures could be attributed to increased enzyme activity and cellular processes that facilitate quicker re-expansion of the anemone's body. However, it is crucial to consider that while short-term exposure to higher temperatures may enhance recovery, prolonged exposure could lead to thermal stress, negatively impacting an anemone's health and survival (Putnam & Edmunds, 2011). As a result, further study could consider to carry out a long-term experiment.

The analysis of substrate presence on SRT revealed no significant effect. This finding suggests that, the stability and resource availability provided by mussels do not significantly alter the immediate recovery responses of *Metridium senile*. It is possible that the anemones' inherent physiological mechanisms are robust enough to override the potential benefits of substrate presence in the context of recovery from a brief disturbance. This lack of significant effect aligns with some ecological studies that suggest certain sessile organisms may not always exhibit strong dependencies on their substrates for short-term physiological responses (Sebens, 1983; Helmuth et al., 2002). However, substrate

presence could still play a vital role in other aspects of anemone ecology, such as growth, reproduction, and long-term survival, which were not within the scope of this study.

16 Different genetically distinct morphotypes should be considered. Maskrey et al. (2020) showed that personality is related to anemone morphotypes, where different morphotypes tend to grow on different parts of the shore. Therefore, to minimize the impact caused by variation in personality changes when varying temperatures, it is important to collect anemones with the same morphotype.

Another limitation of our study is that the acclimation time between the swapped trials might be too short, some anemones even did not fully open when we started our second trial after swapping. The short acclimation time might also lead to a carry-over effect or even adaptation to the water squirt stimulation, recognizing it as a non-predatory threat over time. Future studies should consider varying the type of stimulation or 17 introducing more ecologically relevant stressors to better simulate natural predation threats.

52 anemones used in the experiment may not capture the full range of behavioural variability in the population. Increasing the sample size and repeating the experiment with additional sets of anemones would enhance the robustness of the findings. Moreover, testing anemones of varying sizes could help identify size-dependent behaviour. Apart from that, our experiment collected anemones living on the pontoon may not fully represent the natural habitat of *Metridium senile*. Further studies should focus on natural environments to confirm the applicability of these findings.

CONCLUSION

This study demonstrates that temperature changes significantly impact the boldness of *Metridium senile*, with higher temperatures leading to faster SRT. However, there are no significant effects of anemones growing with or without mussels on their immediate physiological response to disturbance. These findings contribute to our understanding of how marine organisms adapt to changing environments, emphasizing the importance of temperature and habitat conditions in ecological studies involving marine organisms.

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Word count: 2079

REFERENCES

- Beckmann, C. & Biro, P.A. (2013) On the Validity of a Single (Boldness) Assay in Personality Research J. Wright (ed.). *Ethology*. 119 (11), 937–947. doi:10.1111/eth.12137.
- Biro, P.A., Beckmann, C. & Stamps, J.A. (2009) Small within-day increases in temperature affects boldness and alters personality in coral reef fish. *Proceedings of the Royal Society B: Biological Sciences*. 277 (1678), 71–77. doi:10.1098/rspb.2009.1346.
- Chomsky, O., Kamenir, Y., Hyams, M., Dubinsky, Z. & Chadwick-Furman, N.E. (2004) Effects of temperature on growth rate and body size in the Mediterranean Sea anemone *Actinia equina*. *Journal of Experimental Marine Biology and Ecology*. 313 (1), 63–73. doi:10.1016/j.jembe.2004.07.017.
- Duarte, C.M., Hendriks, I.E., Moore, T.S., Olsen, Y.S., Steckbauer, A., Ramajo, L., Carstensen, J., Trotter, J.A. & McCulloch, M. (2013) Is Ocean Acidification an Open-Ocean Syndrome? Understanding Anthropogenic Impacts on Seawater pH. *Estuaries and Coasts*. 36 (2), 221–236. doi:10.1007/s12237-013-9594-3.
- Fitt, W.K., McFarland, F.K., Warner, M.E. & Chilcoat, G.C. (2000) Seasonal patterns of tissue biomass and densities of symbiotic dinoflagellates in reef corals and relation to coral bleaching. *Limnology and Oceanography*. 45 (3), 677–685. doi:10.4319/lo.2000.45.3.0677.
- Forsatkar, M.N., Nematollahi, M.A., Biro, P.A. & Beckmann, C. (2016) Individual boldness traits influenced by temperature in male Siamese fighting fish. *Physiology & Behavior*. 165, 267–272. doi:10.1016/j.physbeh.2016.08.007.
- Griffiths, R.J. (1977) Thermal Stress and the Biology of *Actinia Equina* L. (Anthozoa). *Journal of Experimental Marine Biology and Ecology*. 27 (1969), 141–154.
- Helmuth, B., Harley, C.D.G., Halpin, P.M., O'Donnell, M., Hofmann, G.E. & Blanchette, C.A. (2002) Climate Change and Latitudinal Patterns of Intertidal Thermal Stress. *Science*. 298 (5595), 1015–1017. doi:10.1126/science.1076814.
- Jones, C., Lawton, J. & Schachak, M. (1994) Organisms as ecosystem engineers. *Oikos*. 69, 373–386. doi:10.2307/3545850.
- Kroeker, K.J., Kordas, R.L., Crim, R.N. & Singh, G.G. (2010) Meta-analysis reveals negative yet variable effects of ocean acidification on marine organisms. *Ecology Letters*. 13 (11), 1419–1434. doi:10.1111/j.1461-0248.2010.01518.x.
- Maskrey, D.K. (2021) *The Behavioural, Metabolic and Proteomic Effects of Temperature Stress on Bold and Shy Beadlet Anemones (Actinia Equina)*. Ph.D. England, The University of Liverpool (United Kingdom).
<https://www.proquest.com/docview/2665126818/abstract/58321A3CA5E643B2PQ/1>.
- Maskrey, D.K., Killen, S.S., Sneddon, L.U., Arnold, K.E., Wolfenden, D.C.C. & Thomson, J.S. (2024) Differential metabolic responses in bold and shy sea anemones during a simulated heatwave. *Journal of Experimental Biology*. 227 (3), jeb244662. doi:10.1242/jeb.244662.

- Maskrey, D.K., Sneddon, L.U., Arnold, K.E., Wolfenden, D.C.C. & Thomson, J.S. (2021) Temperature-driven changes in behavioural unpredictability and personality in the beadlet sea anemone, *Actinia equina*. *Animal Behaviour*. 181, 13–27. doi:10.1016/j.anbehav.2021.08.022.
- Maskrey, D.K., Sneddon, L.U., Arnold, K.E., Wolfenden, D.C.C. & Thomson, J.S. (2020) The impact of personality, morphotype and shore height on temperature-mediated behavioural responses in the beadlet anemone *Actinia equina* D. Ardia (ed.). *Journal of Animal Ecology*. 89 (10), 2311–2324. doi:10.1111/1365-2656.13301.
- V. Masson-Delmotte, P. Zhai, A. Pirani, S.L. Connors, C. Péan, et al. (eds.) (2021) *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge, United Kingdom and New York, NY, USA, Cambridge University Press. doi:10.1017/9781009157896.
- Ove, H.-G. & John, F.B. (2010) *The Impact of Climate Change on the World's Marine Ecosystems* | *Science*. 2010. https://www.science.org/doi/full/10.1126/science.1189930?casa_token=l0LLk5-U7RgAAAAA%3A8nbFkgFvnaQ9jnGz9SkQJu-JTNiiOBpGjpUP5bRjybGq-etVjMrnWGWZ9Aprj4sjJvb1F2FrZKn_R1Q [Accessed: 14 June 2024].
- Putnam, H.M. & Edmunds, P.J. (2011) The physiological response of reef corals to diel fluctuations in seawater temperature. *Journal of Experimental Marine Biology and Ecology*. 396 (2), 216–223. doi:10.1016/j.jembe.2010.10.026.
- Rudin, F.S. & Briffa, M. (2011) Is boldness a resource-holding potential trait? Fighting prowess and changes in startle response in the sea anemone, *Actinia equina*. *Proceedings of the Royal Society B: Biological Sciences*. 279 (1735), 1904–1910. doi:10.1098/rspb.2011.2418.
- Sebens, K.P. (1983) The larval and juvenile ecology of the temperate octocoral *Alcyonium siderium* Verrill. I. Substratum selection by benthic Larvae. *Journal of Experimental Marine Biology and Ecology*. 71 (1), 73–89. doi:10.1016/0022-0981(83)90105-3.
- Sebens, K.P. (1984) Water flow and coral colony size: Interhabitat comparisons of the octocoral *Alcyonium siderium*. *Proceedings of the National Academy of Sciences*. 81 (17), 5473–5477. doi:10.1073/pnas.81.17.5473.
- Shick, J.M. (2012) Google-Books-ID: RZDpCAAQBAJ. *A Functional Biology of Sea Anemones*. Springer Science & Business Media.

How temperature and substrate affect anemone boldness.pdf

GRADEMARK REPORT

FINAL GRADE

GENERAL COMMENTS

58/100

PAGE 1

Text Comment. great

PAGE 2

PAGE 3

Text Comment. start with the background/problem before moving on to the study

Text Comment. Boldness, measured as



Comment 1

they do!



Comment 2

do they? to what extent?

PAGE 4



Comment 3

expand on this to tell is how. Make sure you define boldness as well



Comment 4

omit this to allow expansion of the technical concepts from the previous sentence

Text Comment. good to see, but as you can see the reader wants this info earlier if they are a non expert

Text Comment. good

PAGE 5

Text Comment. water temp?



Comment 5

good, so you should include this in the model as we know that size may affect personality



Comment 6

how- what about the plate and tanks you put them in?



Comment 7

why these and how do they relate to the anemones preferred temp range?

Text Comment. repeatability of SRT to test for boldness personality?



Comment 8

which?

PAGE 6

Text Comment. what about the substrate analysis?



Comment 9

you didn;t run any models

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Text Comment. good

Text Comment. p-value is usually the final column

PAGE 7



Comment 12

yes but do they also highest metabolic rate at the lower temp. This- explain what it shows- it in more detail

Text Comment. *italics*



Comment 13

is that likely?

Text Comment. *repeated*



Comment 14

Not clear what this is saying.



Comment 15

explain this in more detail

PAGE 8



Comment 16

you collected them from the pontoon- what were the differences you noted. Try to back up speculations with fact and relate to your experiment.



Comment 17

such as?? Try not to be vague

PAGE 9

PAGE 10

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YUKI WU

FORMATTING, TITLE & LAYOUT

 well presented and good formatting

ABSTRACTS

 See comments.

INTRODUCTION

 Well laid out but make sure you give the info the reader when they need it and unpack some of those studies to say what they showed in a bit more detail

METHODS

 Brief and lacks some details, so I could not replicate the rearing or the ICC method for example. Some diagrams would be helpful

RESULTS

 Very brief section but summarises the data analysis well and good images. Higher marks would have been given for including size as a covariate in a model.

DISCUSSION

 Always start by revisiting the aim of the experiment and hypotheses as the reader would have forgotten them. Do not start

with 'our findings. You have conflated boldness and metabolic activity as you did not test for repeatability. Also aa correlation test between SRT and low temp and SRT at high temp would be a useful proxy for you given your design did not include repeated measures. See further comments in the text

REFERENCES



a good range and well formatted

OVERALL



More time needed on design and stats could have been more sophisticated to use all the data you had eg size and to investigate a possible correlation to make the link to boldness more convincing. Try not to be so vague in your phrasing and when discussin g papers, make sure you real unpack what they did eg Maskrey 2024.